Experiment 12

Aim

To develop a mobile application that enables the user to send an email through the app by entering the recipient's email address, subject, and message content using Android's built-in email capabilities.

Procedure

1. Open Android Studio and create a new project using **EmptyViewsActivity**.
2. Design the user interface using:
   - **EditText** for recipient email
   - **EditText** for subject
   - **EditText** for message content
   - **Button** to send email
3. Assign unique IDs to all UI components in the XML file.
4. Open `MainActivity.java`.
5. Link UI components using findViewById().
6. Create an **Intent** using Intent.ACTION_SEND.
7. Set the MIME type as "message/rfc822" to target email apps.
8. Pass data using:
   - `Intent.EXTRA_EMAIL`
   - `Intent.EXTRA_SUBJECT`
   - `Intent.EXTRA_TEXT`
9. Use startActivity() with chooser to open available email apps.
10. Handle cases where no email app is installed.
11. Run the application and test functionality.

Code

XML Layout (activity_main.xml)

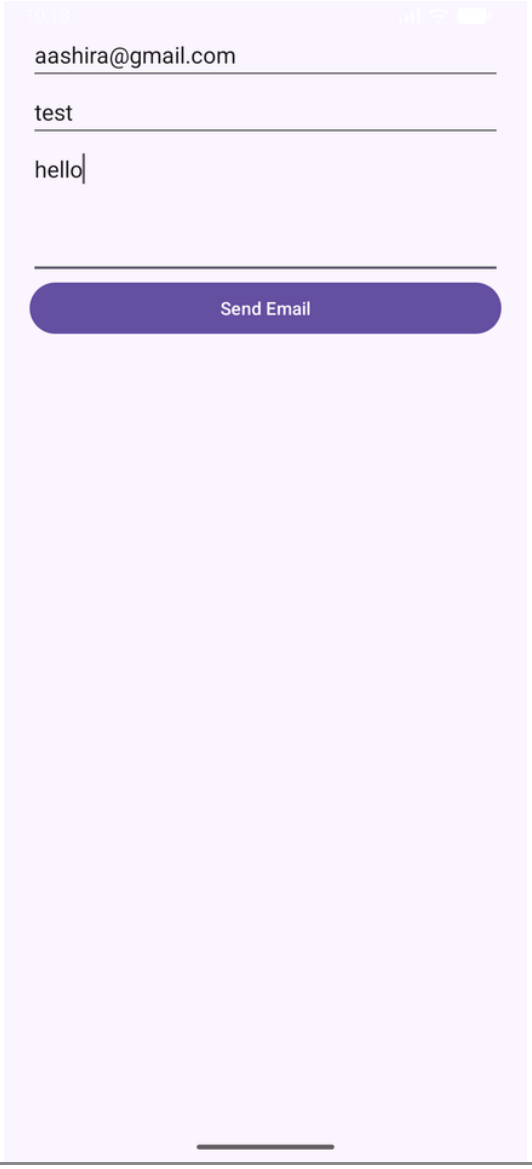
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/email"
        android:hint="Recipient Email"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/subject"
        android:hint="Subject"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/message"
        android:hint="Message"
        android:minLines="4"
        android:gravity="top"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/sendBtn"
        android:text="Send                Email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>
```

Java Code (MainActivity.java)

```Java
package com.example.experiment_12;


import android.content.Intent;
import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText email, subject, message;
    Button sendBtn;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        email    =    findViewById(R.id.email);
        subject  =  findViewById(R.id.subject);
        message  =  findViewById(R.id.message);
        sendBtn = findViewById(R.id.sendBtn);

        sendBtn.setOnClickListener(v -> {

            String   to   =   email.getText().toString();
            String  sub  =  subject.getText().toString();
            String msg = message.getText().toString();

            if (to.isEmpty() || sub.isEmpty() || msg.isEmpty()){
                Toast.makeText(this, "Fill all fields",
Toast.LENGTH_SHORT).show();
                return;
            }

            Intent intent = new Intent(Intent.ACTION_SEND);
```

```
            intent.setType("message/rfc822");

            intent.putExtra(Intent.EXTRA_EMAIL, new String[]{to});
            intent.putExtra(Intent.EXTRA_SUBJECT, sub);
            intent.putExtra(Intent.EXTRA_TEXT, msg);

            try{
                startActivity(Intent.createChooser(intent, "Choose
Email App"));
            }catch(Exception e) {
                Toast.makeText(this, "No email app found",
Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Output



Result

Thus, an Android application was successfully developed to send emails using Android's built-in email functionality by passing recipient address, subject, and message through an intent.